We start with the features and how the application behaves and then go over the technical decisions. I conclude this chapter with some important next steps to your learning path for building robust, full-stack, AI-powered web applications.

11.1 Overview of the application

Our RAG web app allows users to create, manage, and query knowledge bases composed of multiple documents (PDFs and DOCX supported). Each user can securely authenticate, create or delete knowledge bases, upload new documents to them, and chat with an AI assistant that answers questions based on the knowledge within the selected knowledge base.

This is useful when you want to build, for example, a corporate knowledge base for your employees or customers to chat with using a conversational agent.

11.1.1 Key features

In the first iteration of the project, I provide a basic set of features that accomplish the task. Along with the methods and techniques we already know, we'll add some new, unique features.

AUTHENTICATION WITH CLERK

User authentication happens with Clerk.js, like the previous applications. Clerk.js is a solid choice since it provides secure and fine-grained control of the user logins and user information storage. We require all users to authenticate to the application before they gain access to the chat interface.

MAIN DASHBOARD PAGE

Once logged in, users begin their experience by navigating to the main dashboard page. This page contains an overview of the existing knowledge base and useful links and shortcuts to access key application features. Figure 11.1 shows the UI of the main dashboard page.

If you have existing knowledge bases created, they will be listed here, and you can begin chatting. Clicking View All Links when you don't have a knowledge base created will prompt the user to create one instead. Figure 11.2 shows the UI part of this as well.

CREATING A NEW KNOWLEDGE BASE

To create a new knowledge base, click the New Knowledge Base button. This action takes you to a form where you can configure your knowledge base by providing the necessary details. The form requires the following inputs:

- *Name of the knowledge base*
- *Description*—An optional field where you can add a brief explanation or purpose of the knowledge base

Figure 11.3 shows a screenshot of the form with the fields completed.